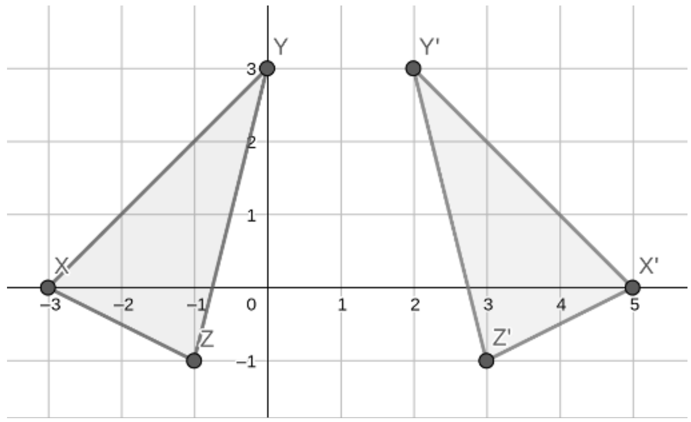
Grade 8 Name_Answer Key_________
Unit 9 Date____________________
Lesson 4 Practice Period__________________

On the coordinate plane shown, $\triangle XYZ$ is reflected to create $\triangle X'Y'Z'$.



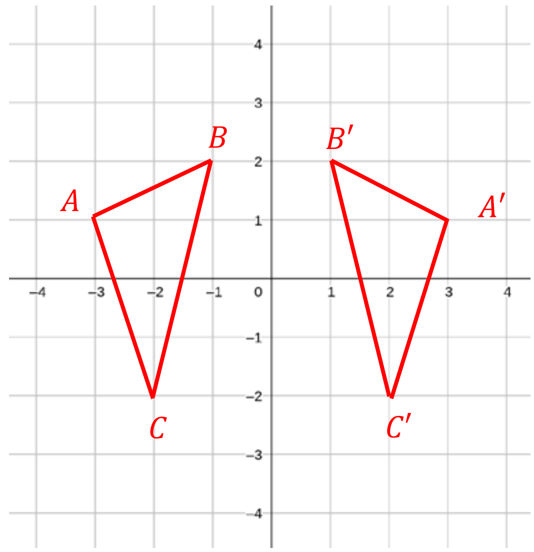
1. What is the line of reflection?
$x = 1$

Using the coordinate plane in question 1, complete the table with the coordinates of each vertex.

Coordinates		Coordinates	
2. Point X	$(-3,0)$	**Point X'**	$(5,0)$
3. Point Y	$(0,3)$	**Point Y'**	$(2,3)$
4. Point Z	$(-1,-1)$	**Point Z'**	$(3,-1)$

5. Describe the relationship between the coordinates of the preimage and the image.
The preimage and image have the same y-coordinates, but different x-coordinates.

6. The vertices of $\triangle ABC$ have their coordinates given in the below table. Plot ABC on the coordinate grid.



Reflect $\triangle ABC$ over the y-axis. Label the image $\triangle A'B'C'$ and complete the table for the image.

Coordinates		Coordinates	
7. Point A	$(-3, 1)$	**Point A'**	(3,1)
8. Point B	$(-1, 2)$	**Point B'**	(1,2)
9. Point C	$(-2, -2)$	**Point C'**	$(2, -2)$

10. The distance from point A to the y-axis and the distance from point A' to the y-axis are

- o the same.
- o not the same.